[INCH-POUND]
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September 5, 1996

COMMERCIAL ITEM DESCRIPTION

TORCHES, OXYACETYLENE, HAND HELD, CUTTING AND WELDING

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers hand held, oxyacetylene torches used for cutting and welding ferrous metals.

1.2 Intended use. The torches covered by this CID are intended for use by the military for field maintenance operations on military equipment.

2. CLASSIFICATION

2.1 Classification. Torches shall be of the following types. The type to be furnished shall be as specified (see 7.2).

Type I - Torch, combination, welding and cutting.
Type II - Torch, cutting.

3. SALIENT CHARACTERISTICS

3.1 General requirements. The torch shall be a hand held oxyacetylene torch, of the positive pressure type, capable of mixing and controlling the flow of gases to the torch tip. The torch shall meet the requirements of CGA E-5 and shall be listed with UL as meeting the requirements of UL123 and marked in accordance with UL labeling requirements.

Beneficial comments, recommendations, additions, deletions, etc. and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3433

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3.2 Type I torch. The type I torch shall include a torch handle, welding tips (as specified in section 3.6), cutting attachments (as specified in section 3.7), and cutting tips (as specified in section 3.8).

3.3 Type II torch. The type II torch shall be designed strictly for cutting and shall be supplied with cutting tips as specified in section 3.8. The cutting head shall be at an angle of 90 ± 2 degrees relative to the centerline of the torch handle.

3.4 Control valves. The preheat control valve shall provide a seal that does not allow leak-through at a pressure of 50 pounds per square inch gage (psig), or greater, when the valve is closed with no more than 20 inch-pounds of torque applied at the knob. The cutting oxygen control valve shall not leak externally at a pressure of 50 psig, or greater, when the valve is open or closed.

3.5 Gas inlets. Each inlet shall have either a reverse flow check valve or flashback arrestor that meets the requirements of CGA E-2. A reverse flow check valve or flashback arrestor is not required if it is built into the torch handle and meets the requirements of CGA E-2. Hose connections shall be at the rear of the torch handle and shall meet the requirements for threaded connections 022 and 023 of CGA E-1. Each inlet shall be clearly and permanently marked to identify the fuel/acetylene and oxygen lines.

3.6 Welding tips. Welding tips for use with the type I torch shall be of the goose-neck design and shall be provided with a mixer. If welding tips are not integral with their mixer, the tips shall be threaded externally to provide a positive, gas tight, metal-to-metal fit. The tips shall provide a long bulbous flame symmetrical with the axis of the tip end. Unless otherwise specified (see 7.2), the type I torch shall be provided with three welding tips and shall be of the drill sizes shown in table I. Tips shall be legibly and permanently marked for size according to the manufacturer's standard practice.

TABLE I. Welding tip sizes.

Size	Drill size no. (Tip orifice)	Welding capacity, in. (Mild steel)
1	72 - 70	3/64
2	58 - 56	1/8
3	40 - 35	3/4

3.7 Cutting attachment. The type I cutting attachment shall consist of a mating part and retaining nut for mounting to the torch handle; gas tubes for carrying gases to the tip; a torch head; a tip retaining nut; a preheat oxygen control valve; and a lever-operated valve to control oxygen flow. The cutting head shall be at an angle of 90 ± 2 degrees relative to the centerline of the torch handle. The preheat oxygen valve and the cutting oxygen valve are not required

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on the cutting attachment if these functions are provided on the torch handle. The cutting attachment shall be mountable onto the torch handle by hand tightening only and shall provide a gas-tight fit.

3.8 Cutting tips. The cutting tips shall be of a one-piece design for use with acetylene and shall be configured with at least four preheat orifices uniformly distributed around the oxygen orifice. The oxygen orifice shall provide a stream of oxygen, which travels down the centerline of the preheat flame group and shall not deviate from the centerline by more than two degrees. Unless otherwise specified (see 7.2), the type I torch shall be provided with tip sizes one through five, and the type II torch shall be provided with tip sizes one through eight as shown in table II. Tips shall be legibly and permanently marked for size according to the manufacturer's standard practice.

TABLE II. Cutting tip sizes.

Size	Drill size no. (Oxygen orifice)	Cutting capacity, in. (Mild steel)
1	71 -68	1/8
2	66 - 62	1/4
3	59 - 55	1/2
4	56 - 54	1
5	53 - 51	1-1/2
6	50 - 49	2
7	45 - 44	3
8	41 - 38	6

3.9 Wrench. The torch shall be supplied with a wrench(s) having suitable openings for tip nuts, hose connections, and packing nuts where adjustments are required. The wrench(s) shall be inherently corrosion resistant or shall be plated or treated for corrosion resistance.

3.10 Performance.

3.10.1 Oxygen flow rate. The type I torch, with the largest cutting tip (see table I) installed in the cutting attachment, shall have a cutting oxygen flow rate of 450 cubic feet per hour (cfh) or more at a torch oxygen inlet pressure not exceeding 90 psig. The type II torch, with its largest cutting tip (see table II) installed, shall have a cutting oxygen flow rate of 850 cfh or more at a torch oxygen inlet pressure not exceeding 85 psig.

3.10.2 Acetylene flow rate. The type I torch, with the largest cutting tip (see table I) installed in the cutting attachment and the preheat flame adjusted to neutral, shall have an acetylene flow rate of at least 25 cfh at a torch acetylene inlet pressure not exceeding 10 psig. The type II torch, with the largest cutting tip (see table II) installed and the preheat flame adjusted to

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neutral, shall have an acetylene flow rate of at least 40 cfh at a torch acetylene inlet pressure not exceeding 10 psig.

3.10.3 Leakage. Joints, valves, o-ring seals, metal-to-metal mating seals, gas passages, connections, and glands shall be capable of sustaining gas pressures of at least 50 psig internally and externally. Valves shall sustain the gas pressures without leakage sufficient for formation of a bubble when submerged in clean water for 10 seconds, both when the valves are closed and when the valves are open to gas passage. The cutting tip seats shall not allow gas leakage at operating pressures, or higher, when tightened with no more than 20 foot-pounds of torque applied at the tip nut.

3.10.4 Backfire and flashback. The type I and type II torches shall withstand five backfires (see 7. 4), at a rate of a least one per second with no flashback (see 7.4), when the oxygen and acetylene are set at the manufacturer's recommended pressures and the preheat flame is adjusted to neutral.

4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered or environmentally preferable materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be met during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

6. PACKAGING

6.1 Packaging requirements. Preservation, packing, and marking shall be as specified (see 7.2).

7. NOTES

7.1 Source of documents.

7.1.1 Government documents. Copies of Military and Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

Underwriters Laboratories (UL)

UL-123 UL Standard for Safety Oxy-Fuel Gas Torches

Applications for copies should be sent to Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062

Compressed Gas Association (CGA)

CGA E-1 Standard Connections For Regulator Outlets, Torches, and Fitted Hose For Welding and Cutting Equipment.

CGA E-2 Hose Line Check Valve Standards For Welding and Cutting

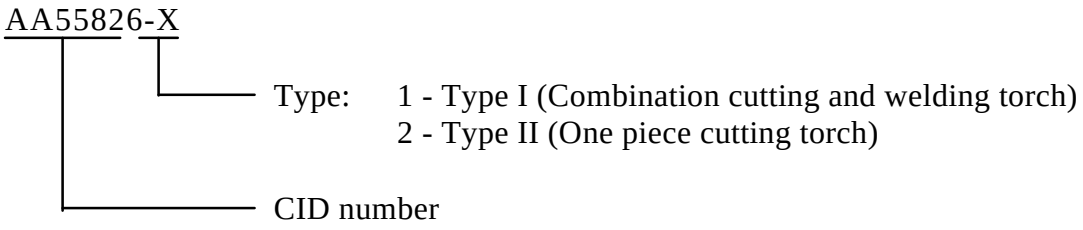
CGA E-5 Torch Standard For Welding and Cutting

Applications for copies should be sent to Compressed Gas Association, Inc., 1235 Jefferson Davis Highway, Arlington, Virginia, 22202.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type of torch required (see 2.1).
c. Welding tips, if different (see 3.6).
d. Cutting tips, if different (see 3.8).
e. Packaging requirements (see 6.1).

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



7.4 Definitions. Terms used in this document are defined as follows:

a. Backfire. The explosion of gases due to a momentary retrogression of the flame into the torch tip, usually caused by snuffing the burning tip against any surface or by an

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imbalance of gases. A typical backfire (a loud “pop”) results in immediate, automatic extinguishing of the flame and, if an ignition source exists outside of the tip, the mixed gases then promptly re-light and burn properly.

b. Flashback. A retrogression of the flame into the torch, as far back as the mixing chamber, and a sustained burning of gases inside the torch or tip.

7.5 National stock numbers (NSNs). The following NSNs correspond to this CID:

3433-00-273-2552	3433-01-075-2135
3433-00-294-6743	3433-01-093-8012
3433-00-876-6996	3433-01-150-2883
3433-01-070-9838	3433-01-158-6597

7.6 Known sources. The following is a list of sources known to meet the requirements of this CID. The list may not be indicative of all possible sources.

Thermal Dynamics Corp
Uniweld Products Inc.
Victor Equipment

MILITARY INTERESTS:

Custodians
Army - AL
Air Force - 99
Navy - YD1

Reviewers
Air Force - 84
Navy - MC
Army - AV

PREPARING ACTIVITY:

DLA - GS

(Project 3433-0136)

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